



Predicting Suicide Deaths in Shandong Province, China.

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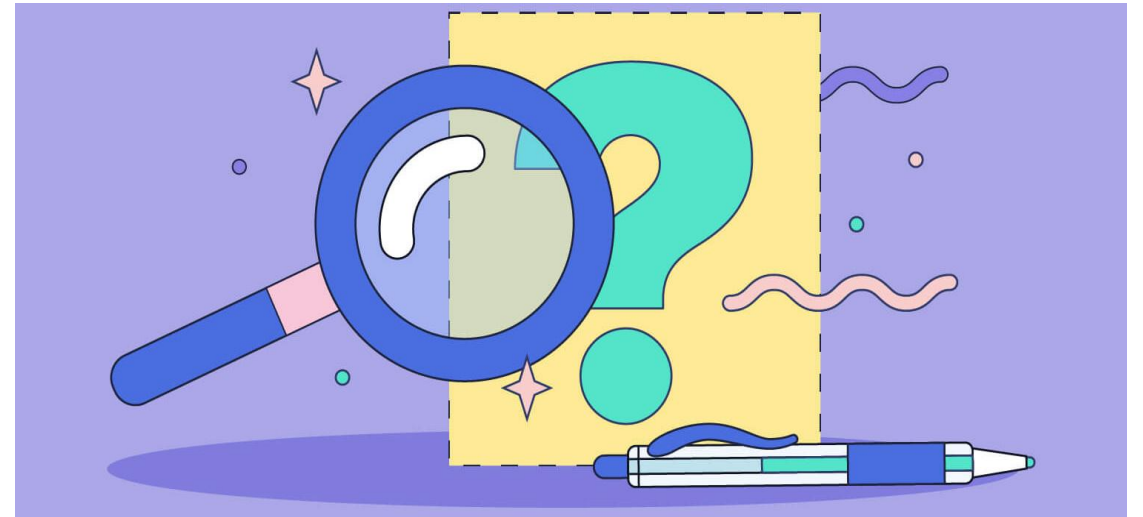
Background

- Definition of Suicide
 - Is the act of voluntarily taking one's own life.
- ~700,000 yearly suicide deaths worldwide
- China's rate?
 - Higher for Females
 - Higher in rural



Questions

- 1. Does sex and location of residence, among other factors (age, highest education level, and method of attempt) predict suicide deaths?
- 2. Do sex and location simultaneously affect the odds of suicide deaths?



Methods

- Data: Shandong Province, China
- Validation techniques used:
 - Variable selection
 - Goodness-of-fit test
 - Residual diagnostic plots
 - ROC and confusion matrix
- Confidence level: 0.05

Variable Breakdown		
Variable	Type	Categories
Suicide Death (Outcome)	Binary (meaning 2 possible outcomes)	Died, Survived
Location of Residence (Rural) (Predictor)	Categorical (3 options)	Yes (Urban), No (Rural), Unknown
Age (Predictor)	Numeric	N/A
Sex (Predictor)	Categorical (2 options)	Female, Male
Method of Attempt (Predictor)	Categorical (8 options)	Cutting, Hanging, Jumping, Other, Other poison, Pesticide, Poison with type unspecified, Unspecified
Highest Education Level (Predictor)	Categorical (5 options)	Illiterate, Primary, Secondary, Tertiary, unknown
Reference levels: Location (Urban), Sex (Male), Method of Attempt (Unspecified), Highest Education Level (Unknown)		

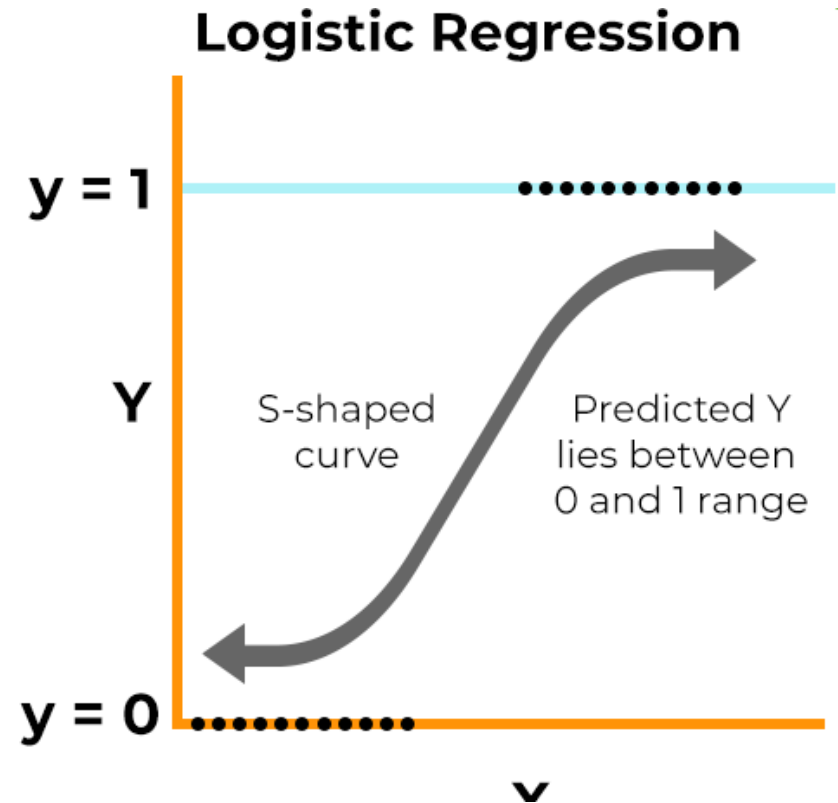
Validated Logistic Model

$$\ln\left(\frac{\text{probability of Suicide Death}}{1-\text{probability of Suicide Death}}\right) =$$

Location of Residence(Predictor) + Age(Predictor) +

Sex(Predictor) + Method of Attempt(Predictor) +

Highest Education Level(Predictor)



Results

EFFECTS OF THE PREDICTORS ON THE RESPONSE

Predictor	P-Value	Significant?
Location of Residence: Rural	~ 0	Yes
Location of Residence: Unknown	~ 0	Yes
Age	~ 0	Yes
Sex: Female	0.003	Yes
Method of Attempt: Cutting	0.038	Yes
Method of Attempt: Hanging	~ 0	Yes
Method of Attempt: Jumping	~ 0	Yes
Method of Attempt: Other	~ 0	Yes
Method of Attempt: Other poison	0.748	No
Method of Attempt: Pesticide	~ 0	Yes
Method of Attempt: Poison with type unspecified	0.408	No
Highest Education Level: Illiterate	~ 0	Yes
Highest Education Level: Primary	~ 0	Yes
Highest Education Level: Secondary	0.013	Yes
Highest Education Level: Tertiary	~ 0	Yes

Predictor	Coefficients
Location of Residence: Rural	3.961
Location of Residence: Unknown	4.004
Age	0.019
Sex: Female	0.332
Method of Attempt: Cutting	3.474
Method of Attempt: Hanging	2.714
Method of Attempt: Jumping	1.164
Method of Attempt: Other	2.994
Method of Attempt: Pesticide	5.964
Highest Education Level: Illiterate	6.312
Highest Education Level: Primary	0.225
Highest Education Level: Secondary	2.586
Highest Education Level: Tertiary	-0.722

Results: Question 1

1. Yes, sex and location of residence, age, highest education level, and method of attempt do predict suicide death.

- No: Method (Other poison); Method (poison with type unspecified)



Results: Question 2

2. Do sex and location simultaneously affect the odds of suicide deaths?

- No.



Conclusion

- These findings will benefit people in a significant way.
 - Help at risk groups
- Woman AND rural?
or....woman OR rural?
 - More research

